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Sea Lice (*Lepeophtheirus salmonis*) Harbour Putative Fish Pathogens: Insights From Illumina and Nanopore Sequencing

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ABSTRACT

Ectoparasites that penetrate host skin can act as biological or mechanical vectors for pathogens and, in some cases, serve as reservoirs. Crustacean ectoparasites of fish are potential vectors of pathogens, which is especially relevant for obligate pathogens (e.g., *Aeromonas salmonicida*) with limited seawater survival. Sea lice (*Lepeophtheirus salmonis*), affecting Atlantic salmon, cause dermal damage and can facilitate secondary infections, resulting in economic losses. While the physical impact of sea lice is well known, their role in pathogen transmission is less clear. The gut bacterial microbiome of lice collected over four months from a salmon farm in Ireland was analysed using Illumina MiSeq and Oxford Nanopore Technologies (ONT) PromethION sequencing for comparison. Illumina and Nanopore sequencing identified 15 and 24 genera of known fish pathogens, respectively. Moreover, Nanopore data revealed up to 15 putative pathogenic species, including *Tenacibaculum maritimum*, *T. dicentrarchi* and *Vibrio anguillarum*, causative agents of tenacibaculosis and vibriosis. The results of this study provide a gut bacterial microbiome characterisation of *L. salmonis* in a commercial aquaculture setting and demonstrate the potential of sea lice to act as pathogen vectors or reservoirs. These findings have important implications for pathogen surveillance, management, and prevention strategies in salmon aquaculture.

1 | Introduction

Ectoparasites that penetrate the epidermis and feed on multiple hosts throughout their lifecycle have the potential to act as vectors of pathogens (Cusack and Cone 1986; Wall 2007). They may function as mechanical vectors, passively carrying pathogens on their surface without facilitating pathogen replication, or as biological vectors, in which the pathogen multiplies or develops before transmission (Sarwar 2015). In addition, some ectoparasites can also serve as reservoirs, harbouring pathogens long-term and acting as a continuous source of infection for susceptible populations (Haydon et al. 2002; Ashford 2003).

Crustacean ectoparasites of fish have been identified as potential vectors of various pathogens (Cusack and Cone 1986; Mackenzie and Jeggo 2013; Novak et al. 2016; Bass et al. 2021). This mode of transmission may be particularly relevant for obligate pathogens such as *Aeromonas salmonicida*, infectious pancreatic necrosis virus, and viral haemorrhagic septicaemia virus, which have limited survival in seawater (Cusack and Cone 1986). However, many pathogens do not rely exclusively on vectors and, at high abundance levels, can also infect hosts directly through dissemination in seawater (Cusack and Cone 1986). Understanding pathogen transmission routes and identifying infection sources are essential for effective disease

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management and outbreak prevention in aquaculture (Haydon et al. 2002; Wilson et al. 2020).

Ectoparasitic sea lice pose a major challenge to Atlantic salmon aquaculture (Jackson et al. 2002; Liu and Bjelland 2014). The most commercially important sea lice species in the North Atlantic is *Lepeophtheirus salmonis*, however, high infection levels of *Caligus elongatus* on salmon farms are also common (Hemmingsen et al. 2020; D'Arcy et al. 2025). By feeding on the mucus, blood, and tissue of hosts, sea lice can compromise fish welfare, causing dermal damage, secondary infections, and, in severe cases, increased mortality (Johnson et al. 2004; Finstad et al. 2000). Sea lice infections on salmon farms lead to substantial economic losses, primarily due to the costs associated with management and treatment (Liu and Bjelland 2014; Torrisen et al. 2013; Overton et al. 2019). Consequently, effective control strategies depend on monitoring programmes that assess sea lice population levels and enable timely interventions (Treasurer and Pope 2000; Westcott et al. 2004; Jeong et al. 2023). Despite the substantial economic impact of sea lice, their role in transmitting salmon pathogens remains largely unclear (Nylund et al. 1991; Jakob et al. 2011; Novak et al. 2016).

To date, only three studies have investigated the gut bacterial microbiome of sea lice. Two of these focused on *Caligus rogercresseyi* (Gonçalves et al. 2020; Morales-Rivera et al. 2022), while one examined *Lepeophtheirus salmonis* (Dalvin et al. 2020). In the latter study, *L. salmonis* was collected from multiple locations across the Atlantic and Pacific Oceans; although diverse microbial communities were described in the gut bacterial microbiome, no pathogens were detected (Dalvin et al. 2020). *C. rogercresseyi* is primarily distributed in the Southern Hemisphere (Gonçalves et al. 2020), whereas *L. salmonis* is considered the most commercially important sea lice species in the Northern Hemisphere, particularly within the Atlantic region (Skern-Mauritzen et al. 2014; Jackson et al. 2002). Beyond their differences in geographic distributions, these two genera also vary in morphology and life cycle. Notably, *L. salmonis* is relatively host-specific, infecting salmonids, whereas *Caligus* species have a broader host range (Hemmingsen et al. 2020). Several studies have detected viral salmonid pathogens in *L. salmonis* (Nylund et al. 1991; Jakob et al. 2011; Novak et al. 2016). Studies that have investigated bacterial pathogens in *L. salmonis* have identified *Aeromonas salmonicida* subsp. *salmonicida*, *Pseudomonas fluorescens*, *Tenacibaculum maritimum*, and *Vibrio* spp. (Barker et al. 2009; Novak et al. 2016). Although disease outbreaks have been reported in association with sea lice infections, these are primarily linked to sea lice compromising host resistance to secondary pathogens (Bustos et al. 2011; Barker et al. 2013; Figueroa et al. 2017).

The Illumina MiSeq platform is a well-established tool for microbiome characterisation (Stevens et al. 2023). More recently, the Oxford Nanopore Technologies (ONT) PromethION sequencer has emerged as an alternative approach, offering several advantages, including full-length 16S rRNA gene sequencing, which enhances taxonomic identification accuracy (Edwards et al. 2018). Additionally, Nanopore provides real-time sequencing with minimal equipment requirements, making it particularly suited for field use in remote locations (Edwards et al. 2018; Gonçalves et al. 2020; Stevens et al. 2023). While early Nanopore

models were limited by relatively high error rates (read accuracy ~85%–94%), recent advancements in chemistry and bioinformatics pipelines have greatly improved accuracy to around ~99% (Zhang et al. 2023; Curry et al. 2024).

Understanding the role of sea lice in the life cycle and transmission of salmon pathogens is crucial for effective disease management in Atlantic salmon farms (Haydon et al. 2002; Wilson et al. 2020). This study had three main aims: (1) to characterise the gut bacterial microbiome of sea lice and assess its changes over a four-month period; (2) to investigate the potential of sea lice to function as vectors and/or reservoirs of salmon pathogens; and (3) to compare gut bacterial microbiome analyses using two sequencing platforms: Illumina MiSeq and Oxford Nanopore Technologies PromethION.

2 | Methods

2.1 | Sample Collection

Adult male and female sea lice (*L. salmonis*) were collected from a commercial salmon farm located on the west coast of Ireland (53°15'57.6" N, 9°44'34.8" W) by the Marine Institute, as part of the National Sea Lice Monitoring Programme. Samples selected for this study were collected monthly over a four-month period (2 February, 18 March, 29 April, and 11 May 2022) from a single pen, following the established protocol described by Klimesova et al. (2026). Collected lice were preserved in 70% ethanol and stored at room temperature. The aquaculture site was selected to ensure that fish had been in seawater for a minimum of eight months (as per Gonçalves et al. 2020) and to ensure *L. salmonis* infestation levels were sufficient for the required sampling over the four consecutive months.

2.2 | DNA Extraction and 16S rRNA Gene Amplification

For each of the four months (February to May), 60 adult sea lice were collected and combined into five pooled samples of 12 individuals each, with a male to female ratio of 1:1. To reduce contamination from surface-associated bacteria, sea lice were first rinsed with fresh 70% ethanol before dissection. The lice were then dissected to minimise non-gut biomass; however, surface parts remained present in all samples. For each pooled sample, DNA extraction was carried out using two Eppendorf tubes, each containing six adult sea lice (three males and three females). During the extraction process, the contents of the two tubes corresponding to the same pooled sample were combined when transferred to the spin column, resulting in one final DNA sample representing 12 individuals. In total, this produced 20 pooled samples across the four months. Samples were homogenised using mechanical disruption in 360 µL of ATL buffer with 2 µL of antifoam (Reagent DX, Qiagen). Four ceramic beads (2.8 mm, Qiagen) were used per tube and each Eppendorf tube (2 mL) was wrapped in parafilm. The bead beater (Bead Mill 4, Fisher Scientific, USA) was used at 5 m/s for 15 s, three times with a minute in between rounds. DNA was extracted using the Qiagen DNeasy Blood and Tissue Kit (Qiagen, USA). The manufacturer's protocol was modified to account for the high sample

biomass, by using double volumes of reagents up to the transfer to the spin columns, extending the lysis step to a three-hour incubation with proteinase K, and eluting the DNA from the spin columns with 70 µL of nuclease-free water. All other steps were performed according to the manufacturer's instructions.

Extracted DNA concentration was quantified using a NanoDrop One/One^c spectrophotometer (Thermo Fisher Scientific, USA) and a Qubit 3.0 Fluorometer (Life Technologies, USA) using the Qubit dsDNA BR Assay Kit (Thermo Fisher Scientific, USA), following the manufacturer's instructions. DNA samples were normalised to 200 ng/µl and the presence of bacterial DNA was evaluated by amplification of the 16S rRNA gene. The amplification was performed by PCR with a reaction volume of 50 µL, using the primers 8–27F: 5'-AGAGTTTGATCCTGGCTCAG-3' and 536 R: 5'-GWATTACCGCGGCKGCTG-3' and Taq DNA polymerase (Thermo Fisher Scientific, USA), with an initial denaturation step at 95°C for 2 min, followed by 35 cycles of 95°C for 30 s, 58°C for 30 s and 72°C for 45 s, and an extension at 72°C for 5 min. PCR product sizes were verified by electrophoresis on a 1.5% TAE (Tris-acetate-EDTA) agarose gel. Extracted DNA was stored at –20°C until sequencing.

2.3 | Illumina MiSeq and Nanopore PromethION Sequencing

Illumina MiSeq sequencing was performed commercially, Novogene (Cambridge, United Kingdom). The 16S rRNA gene (region V3–V4) was amplified using 0.2 µM of specific primers carrying sample barcodes (341 F: 5'-CCTAYGGGRBGCASCAG-3' and 806 R: 5'-GGACTACNNGGTATCTAAT-3'), 5 µL of Phusion High-Fidelity PCR Master Mix, and 10 ng of template DNA. The initial denaturation step at 98°C for 1 min was followed by 30 cycles of 98°C for 10 s, 50°C for 30 s, and 72°C for 30 s and extension at 72°C for 5 min. The purified samples were quantified, normalised, and then combined in equal proportions. Quantified and quality-checked libraries were pooled and sequenced according to the required data output.

In addition to Illumina sequencing, all samples were also sequenced using the third-generation sequencing platform PromethION (P2 Solo, Oxford Nanopore Technologies) at the Institute of Aquaculture Torre de la Sal (IATS-CSIC) following the approach described in Domingo-Bretón et al. (2024). The full-length V1–V9 region (27–1492 nt) of 16S rRNA gene (F: 5'-AGAGTTTGATCMTGGCTCAG-3' R: 5'-CGGTTACCTTGTTACGACTT-3') was amplified using LongAmp Hot Start Taq 2× Master Mix (New England Biolabs, M0533L) in a total volume of 25 µL, with an initial denaturing step at 95°C for 1 min, followed by 25 cycles of 95°C for 20 s, 52°C for 30 s and 65°C for 2 min, and an extension at 65°C for 5 min in a Touchgene gradient thermal cycler (Techne, United Kingdom). The PCR products were purified with Agencourt AMPure XP beads (Beckman Coulter, USA) according to the manufacturer's instructions using 0.4 beads/sample ratio. The samples were eluted in 30 µL of nuclease-free water. PCR product sizes were verified by electrophoresis on a 1.5% TAE (Tris-acetate-EDTA) agarose gel and subsequently quantified using Quant-iT PicoGreen dsDNA Assay Kit (Thermo Fisher Scientific, USA) according to the manufacturer's instructions.

Nanopore libraries were prepared using the Native Barcoding Kit 96 V14 (SQK-NBD114.96, Oxford Nanopore Technologies) following ONT protocol version NBA_9170_v114_revP_30Jan2025. The 16S rRNA amplicons were end-prepared and ligated for unique barcodes. The barcoded libraries of 20 samples, a positive control (microbial mock community, ZymoBiomics HMW DNA standard), and a negative control were pooled into a single tube and quantified using PicoGreen. Adapter ligation was performed, followed by clean-up using Agencourt AMPure XP beads (Beckman Coulter, USA) and PicoGreen quantification according to the manufacturer's instructions. The primed flow cell FLO-PRO114M was loaded with 50 fmol of library and sequenced on a Nanopore PromethION platform (Oxford Nanopore Technologies, UK). Raw data was acquired using the software MinKNOW v6.4.12 (Oxford Nanopore Technologies). The sequencing run was set to reach approximately 300,000 reads per sample.

2.4 | Bioinformatic Analysis

Demultiplexing was performed on reads obtained from Illumina MiSeq, and adaptors were removed through Cutadapt v3.3. Sequence assembly paired-end reads were merged using FLASH v1.2.11 (Magoc and Salzberg 2011). Quality filtering on the raw tags was performed using the fastp v0.23.1 software (Chen 2025). The tags were compared with the reference database (SILVA database [16S/18S]) to detect chimera sequences that were removed using the vsearch package v2.16.0 (Edgar et al. 2011).

All bioinformatic processing of Nanopore sequencing data was carried out using Ubuntu v24.04.1. Basecalling was conducted on reads obtained from the Nanopore PromethION using the most accurate available model (SUP) with Dorado v1.0.1 (Oxford Nanopore Technologies PLC 2025). Demultiplexing and trimming of adapters and barcodes was performed using Dorado v1.0.1. SamTools v1.19.2 (Danecek et al. 2021) was used for file type conversion (bam to fastq) and quality (min qscore 14), and length (1200–1800 nt) filtering was performed using Chopper v0.10.0 (De Coster and Rademakers 2023).

Quality analysis for both Illumina and Nanopore sequencing data was obtained using NanoPlot v1.44.1 and reads from all samples were taxonomically assigned with minimap2 v2.29-r1283 (Li 2021) using SILVA v138.2 as a reference database. Raw sequence data were lodged in the Sequence Read Archives (SRA) under the Bioproject accession number: PRJNA1372954.

Genera previously assigned from the Nanopore dataset were cross-referenced with a curated list of fish pathogen-associated genera (Austin and Austin 2016; Morales-Rivera et al. 2022). For downstream analyses, only genera represented by ≥ 10 reads across the dataset were retained to minimise the influence of low-abundance noise. Species-level identification was restricted to full-length 16S rRNA gene reads and performed using Emu v3.5.4 (Curry et al. 2022), evaluating the top 50 minimap2 alignments per read. Species proposed by Emu were subsequently examined using local BLASTn v2.14.0 searches, applying a minimum identity threshold of 0.96. Only species consistently supported by both approaches were retained.

2.5 | Statistics and Data Analyses

All global singletons were considered artifacts and therefore excluded from the dataset before analysis. Rarefaction analysis was used to assess sequencing depth for both platforms. Data were normalised by converting read counts to relative abundances of detected genera for each sample. To minimise the influence of low-abundance taxa, genera with an overall relative abundance below 0.03% were eliminated for each sequencing platform. Bray–Curtis dissimilarities between samples were calculated from relative abundance data, and principal coordinate analysis (PCoA) was performed to determine temporal variation of gut bacterial microbiomes. Differences between months and platforms were tested by permutational multivariate analysis of variance, PERMANOVA (Anderson 2017).

Alpha diversity indices (Shannon and Simpson), observed richness, and richness index (ACE) were calculated separately for each month and platform, based on read counts. Normality of the data distribution was tested using the Shapiro–Wilk test (Razali and Wah 2011). Differences between months were tested using one-way analysis of variance (ANOVA) followed by Tukey's honestly significant difference (HSD) post hoc test (Kim 2015) for normally distributed data, or followed by the Kruskal–Wallis test (Kruskal and Wallis 1952) and a post hoc Wilcoxon test (Cuzick 1985) for non-normally distributed data.

All genera detected by both platforms were compared against a curated list of known fish pathogens (Austin and Austin 2016; Morales-Rivera et al. 2022). Genera with consistently higher relative abundance (> 1%) were identified, and temporal changes in abundance were assessed using the Kruskal–Wallis test followed by post hoc Wilcoxon tests.

Abundance data of the core microbiota, defined as genera detected by both sequencing platforms, were used to construct a stochastic model based on a comprehensive Bayesian Network (BN) separately for the Illumina and Nanopore sequencing data. Network construction was performed using the online SAMBA tool (Structure-learning of aquaculture microbiomes using a Bayesian approach; Soriano et al. 2023) following the settings as reported by Moroni et al. (2025). Taxa with zero total counts were removed before network construction. Models were fitted using a zero-inflated negative binomial (ZINB) distribution, and edge strength was evaluated with the Bayesian Information Criterion (BIC) and Mutual Information (MI), using thresholds of 0 and 0.05, respectively, to exclude weak associations. The final networks were clustered using the Leiden community detection algorithm, to detect subgroups of densely connected nodes within the microbiota network (Traag et al. 2019). For each network, all reachable node pairs (including indirect connections) were computed using shortest-path distances to obtain the transitive closure of the graph. Reachability sets from both networks were then compared to identify shared and unique edges. Network similarity was quantified using the Jaccard index based on the union and intersection of reachable pairs, and by calculating the proportion of shared connections relative to each network. Genus-level biomarkers for each month were identified using linear discriminant analysis (LDA) effect size (LefSe), applied separately to the abundance data of each sequencing platform.

The threshold for the LDA score for discriminative features was set at the value of 4.

Statistical significance was determined at $p < 0.05$. All statistical analyses were carried out using the statistical computing environment R version 4.3.3 and RStudio version 2024.04.2 (R Core Team 2024). The package 'ggplot2' (Wickham 2016) was used for visualisation, 'microbiomeMarker' (Cao et al. 2022) was used for LefSe analysis, 'phyloseq' (McMurdie and Holmes 2013) was used for data manipulation and analysis, and 'vegan' (Oksanen et al. 2025) was used for visualisation and data analysis. [Supporting Information](#) was generated using R Markdown (Allaire et al. 2025).

3 | Results

3.1 | Sequencing Outputs

The total reads yielded by Illumina MiSeq and Oxford Nanopore Technologies (ONT) PromethION sequencing of the 20 pooled sea lice samples (five samples per month, Feb–May) were 1,622,493 and 3,991,554, respectively. The mean number of reads per sample was 81,125 for Illumina and 199,578 for Nanopore. Illumina sequencing yielded reads with a mean length of 429 bp and an average Phred quality score of 35.98 (estimated base accuracy 99.97%). Nanopore reads averaged 1482.9 bp with a mean Phred score of 20.8 (accuracy 99.17%). Rarefaction analysis showed that the curves for all samples approached an asymptote, indicating that the sequencing depth was appropriate to describe the diversity of the bacterial community ([Supporting Information](#)).

3.2 | Composition of Sea Lice Gut Bacterial Microbiome

Altogether 11 phyla, 13 classes, and 35 orders were identified by Illumina and 13 phyla, 17 classes, and 39 orders by Nanopore. The most abundant phylum was Pseudomonadota (Figure 1A) with an overall abundance of 76% (ranging from 62% to 88%) detected by Illumina and 82% (ranging from 67% to 91%) detected by Nanopore. This phylum was represented by two classes, Gammaproteobacteria (68% Illumina, 73% Nanopore) and Alphaproteobacteria (9% Illumina, 10% Nanopore; [Supporting Information](#)). At the order level, Pseudomonadota was predominantly represented by the order Cardiobacteriales (Figure 1B), with an overall abundance of 53% and 64% by Illumina and Nanopore, respectively. To a lesser degree Pseudomonadota was also represented by Rhodobacterales, Beggiatoales, and Hyphomicrobiales (Figure 1B). The phylum Bacteroidota had an overall abundance of 17% and 9% detected by Illumina and Nanopore (Figure 1A) and was represented by the single class Bacteroidia ([Supporting Information](#)). At the level of order, Bacteroidota was represented by Flavobacteriales and Chitinophagales. The phyla Verrucomicrobiota, Campylobacterota, Patescibacteria, and Cyanobacteriota were detected with overall low abundance levels ranging from 3.2% to 1.3% for both platforms (Figure 1A). Both platforms detected the same groups to be the most abundant in the lice gut bacterial microbiome,

although the overall detected abundance differed between the two sequencing platforms (Figure 1B).

While minor temporal variation was observed, certain genera were detected only in specific months (26 Illumina, 21 Nanopore) at low abundance (<1.7%). A core gut bacterial

microbiome dominated the samples throughout the four-month period (Supporting Information), accounting for about 99% of the total community. Based on the Bray–Curtis distances between bacterial communities, differences between months were detected forming two clusters (Figure 2): February with March and April with May. These differences were detected

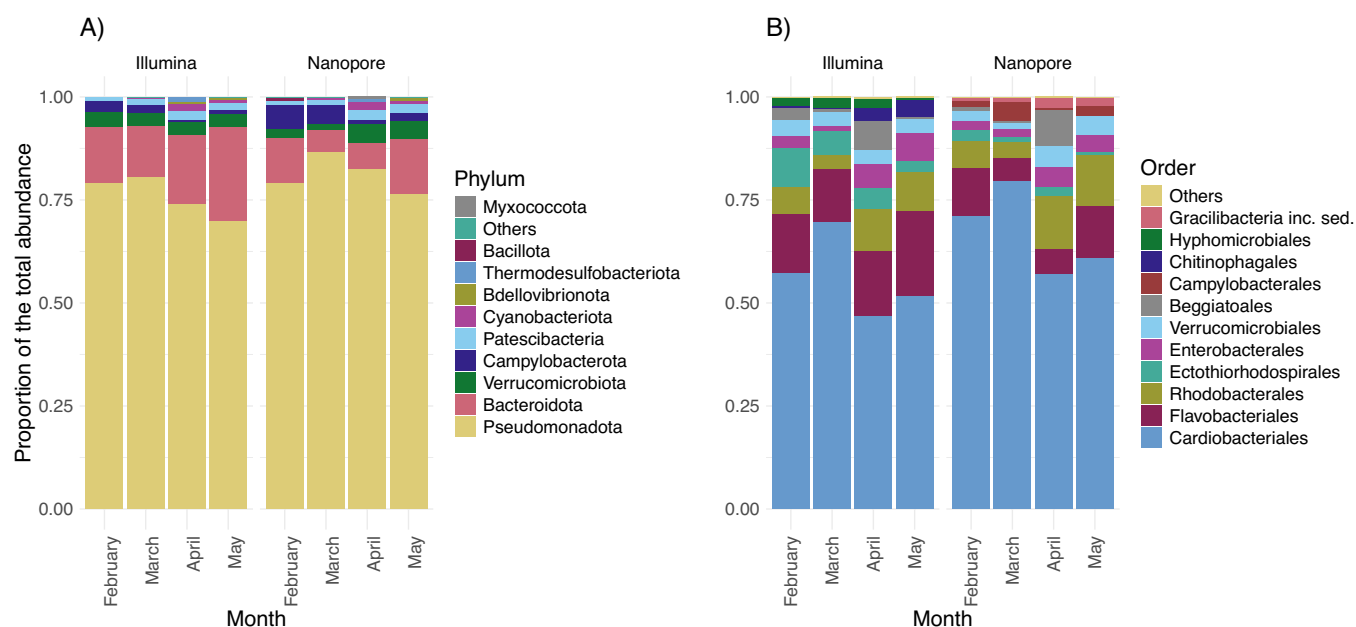


FIGURE 1 | Microbial composition for results obtained by the Illumina and Nanopore sequencing platforms, represented at: (A) phylum and (B) order level for each month separately.

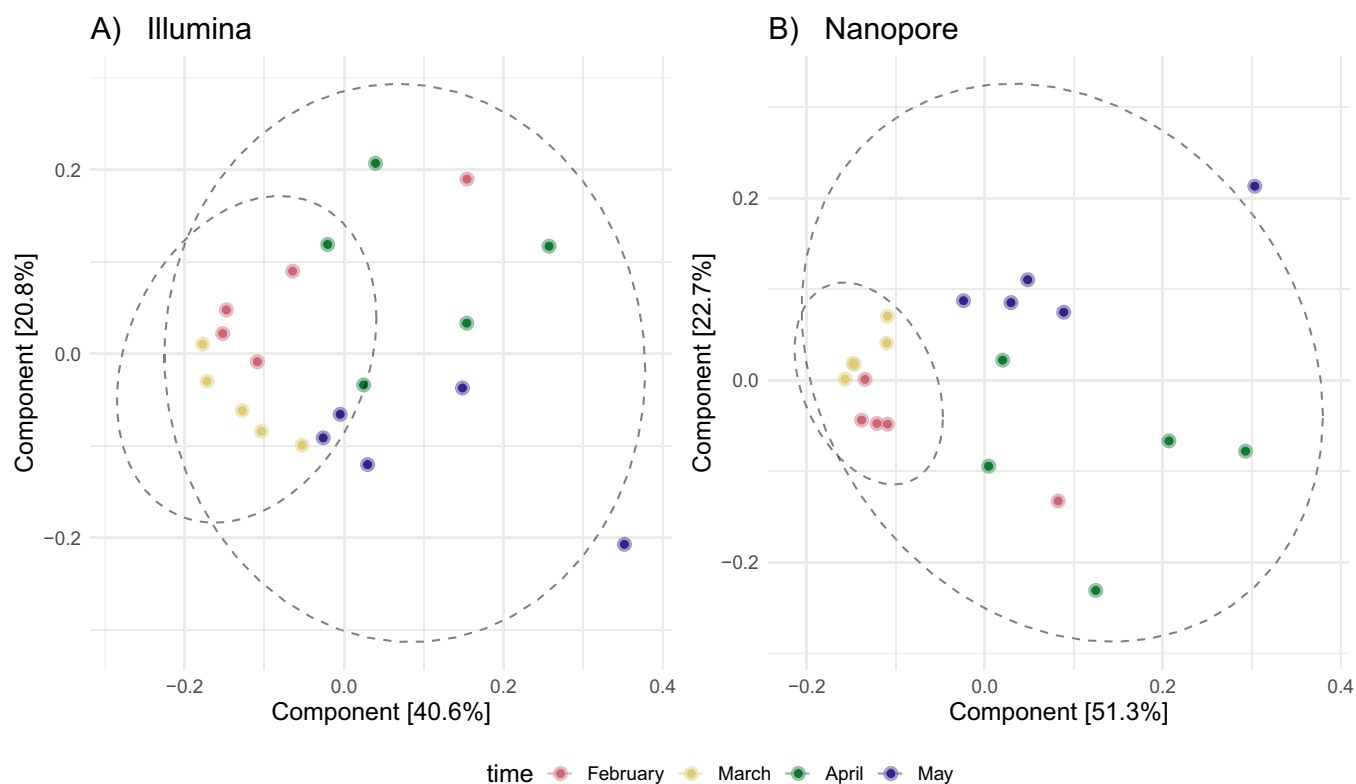


FIGURE 2 | Temporal changes of sea lice gut microbiome analysed by the principal coordinate analysis (PCoA). Confidence ellipses highlight the clustering of months: February with March and April with May.

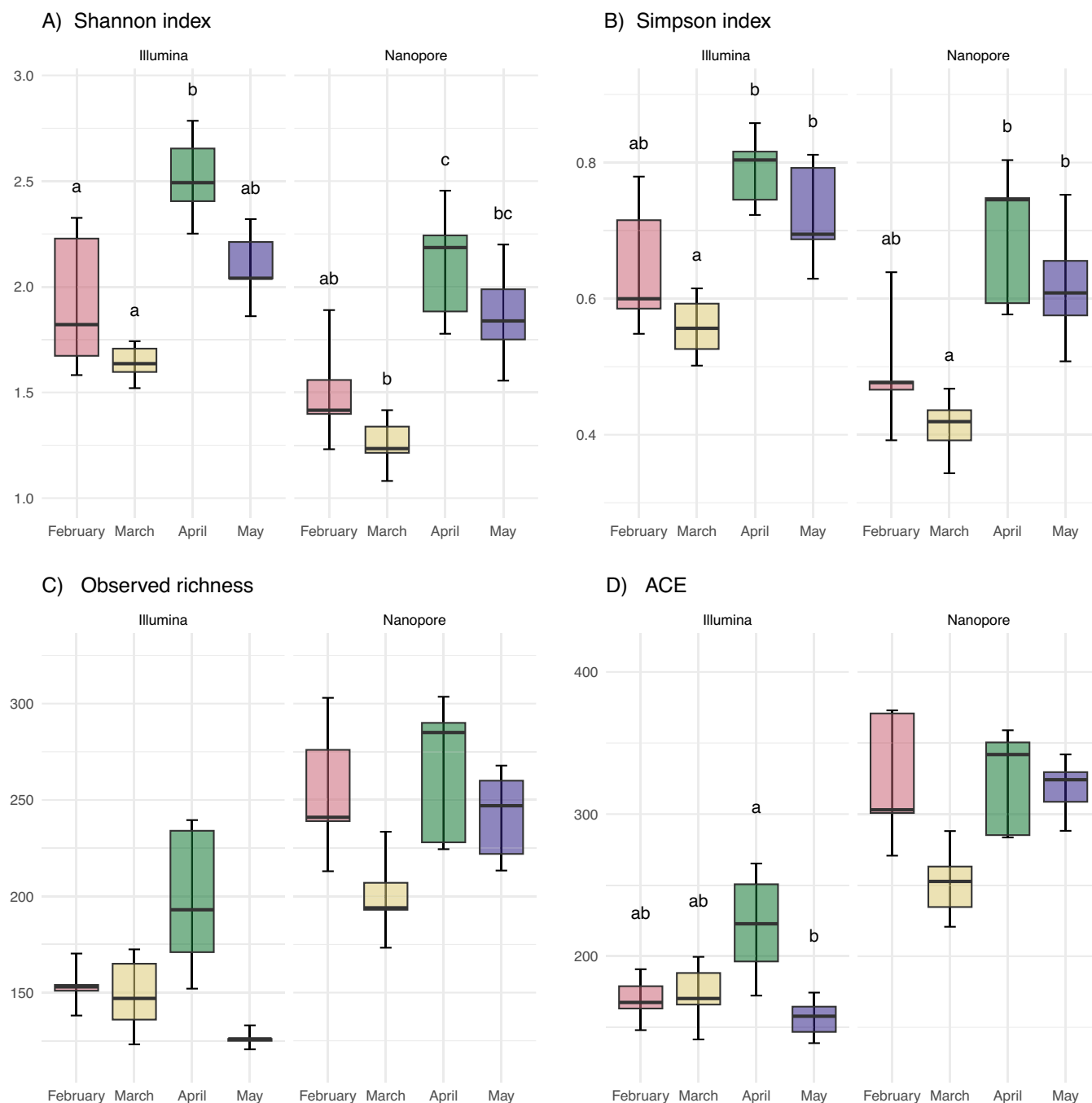


FIGURE 3 | Comparison of sea lice gut microbiome taxa diversity (A, B) and richness (C, D) between the Illumina and Nanopore sequencing: (A) Shannon index, (B) Simpson index, (C) Observed richness, (D) ACE. Statistical differences between months are represented by letters.

in the bacterial communities described by both Illumina and Nanopore, following the same trend. The temporal differences were found to be significant by PERMANOVA for both Illumina ($F = 3.8$, $R^2 = 0.42$, $p < 0.01$) and Nanopore ($F = 6.4$, $R^2 = 0.55$, $p < 0.01$).

Overall, Illumina yielded higher diversity values than Nanopore. Both diversity indices (Shannon and Simpson) showed similar temporal trends. Significant differences in microbial diversity were detected between months, with the highest diversity observed in April and the lowest in March (Figure 3A,B; Supporting Information). The observed microbial richness was higher with Nanopore sequencing than with Illumina (Figure 3).

For Nanopore, richness did not vary significantly across the four months, whereas Illumina data showed the highest richness in April and the lowest in May (Figure 3C,D; Supporting Information).

3.3 | Identification of Potential Fish Pathogens

A total of 24 pathogen-associated genera were detected across both sequencing platforms, with 16 identified by both platforms and a further eight identified by Nanopore (Table 1). Analysis of the Nanopore sequencing data revealed 15 putative pathogenic species within seven genera (Table 1).

TABLE 1 | Number of reads detected for Illumina and Nanopore sequencing platforms for genera containing fish pathogens. Pathogenic fish species from each genus identified by the Nanopore platform.

Genus	Number of reads		Identified pathogenic species: Nanopore
	Illumina	Nanopore	
<i>Acinetobacter</i>	754	291	<i>Acinetobacter johnsonii</i>
<i>Aeromonas</i>	1	29	<i>Aeromonas sobria</i>
<i>Aliivibrio</i>	753	75	<i>Aliivibrio logei</i> , <i>A. wodanis</i>
<i>Arcobacter</i>	17,838	85,966	NA
<i>Carnobacterium</i>	0	32	NA
<i>Chryseobacterium</i>	59	95	NA
<i>Citrobacter</i>	0	160	NA
<i>Enterobacter</i>	0	145	NA
<i>Flavobacterium</i>	51	60	NA
<i>Francisella</i>	707	1163	NA
<i>Halomonas</i>	38	222	NA
<i>Klebsiella</i>	0	27	NA
<i>Moritella</i>	14	14	<i>Moritella marina</i>
<i>Pantoea</i>	10	118	NA
<i>Photobacterium</i>	29	89	NA
<i>Pseudoalteromonas</i>	1606	1239	NA
<i>Pseudomonas</i>	477	849	<i>Pseudomonas fluorescens</i> , <i>P. koreensis</i>
<i>Serratia</i>	14	174	NA
<i>Shewanella</i>	515	899	NA
<i>Streptococcus</i>	7	131	NA
<i>Tenacibaculum</i>	17,152	50,583	<i>Tenacibaculum dicentrarchi</i> , <i>T. maritimum</i> , <i>T. ovolyticum</i> , <i>T. soleae</i>
<i>Vagococcus</i>	0	11	NA
<i>Vibrio</i>	29,992	40,976	<i>Vibrio alginolyticus</i> , <i>V. anguillarum</i> , <i>V. splendidus</i> , <i>V. tapetis</i>
<i>Yersinia</i>	0	25	NA

The genus *Tenacibaculum* was represented by four pathogenic species that showed overall higher relative abundance (0.04%–1.1%) compared to other identified pathogens (Supporting Information). Four species, *T. dicentrarchi*, *T. maritimum*, and *T. ovolyticum*, were detected over the four-month period, whereas *T. soleae* was absent in April. The genus *Vibrio* was also represented by four pathogenic species but at lower overall abundance (0.007%–0.3%). Species occurrence was as follows: *V. splendidus* was detected throughout the four-month period, *V. tapetis* in March, April, and May, *V. alginolyticus* was absent in March and May, and *V. anguillarum* was detected only in March. Other pathogenic species were detected at even lower abundances (0.0003%–0.005%) and typically in a single month: *Aeromonas sobria* in February; *Moritella marina*, *Pseudomonas fluorescens*, and *P. koreensis* in March; *Acinetobacter johnsonii* in April; and *Aliivibrio logei* and *A. wodanis* in May.

The three most abundant pathogen-associated genera identified over the four-month period were *Arcobacter*, *Tenacibaculum*,

and *Vibrio* (Table 1). Significant temporal variation was observed in the abundance of *Tenacibaculum* and *Vibrio*, both showing their lowest abundance in March and highest in May (Figure 4). In contrast, no significant changes were detected in the abundance of *Arcobacter*.

3.4 | Comparison of Illumina and Nanopore Sequencing Platforms

Bayesian networks (BN) were inferred independently from bacterial read counts of Illumina and Nanopore datasets, resulting in distinct yet broadly comparable network structures (Figure 5). The model built for the Nanopore dataset exhibited a total of 82 nodes connected to the main structure, composed of five highly connected clusters, while the Illumina data showed a higher number of nodes (610), but with only 69 taxa directly connected to the network and grouped into seven closely related clusters. The results thus show that regardless of the

input data, the number of nodes and their overall relative abundance (which represents over 90% for both models) within the network-connected bacterial populations were consistent across platforms. Notably, 85% of nodes in the Illumina and 72% in the Nanopore network were shared (Jaccard index=0.64), indicating substantial agreement in the taxa represented. Similar results were also observed for core microbiota taxa, which occupy the majority of the nodes for both models, accounting for a total abundance of 96% and 92% for Illumina and Nanopore BNs, respectively. Consequently, the shared fraction between the two datasets reached a Jaccard index value of 0.63, further supporting the robustness of the structural patterns recovered across sequencing platforms. In contrast, overlap in direct edges was low (9.7% for Illumina and 12.8% for Nanopore), reflecting the expected sensitivity of edge inference to platform-specific differences in read composition and abundance estimates. However, when broader network connectivity was considered, including indirect paths and undirected edges, the number of reachable

pairs has grown substantially, arriving at 4692 and 6642 for the Illumina and Nanopore models, respectively. Considering this aspect, the similarity between networks increased markedly, reaching 73% for Illumina and 52% for Nanopore. Overall, these results indicate that while individual interaction estimates differ between platforms, the BNs captured largely consistent relationships among bacterial taxonomic groups. Most identified genera were common for both platforms (Figure 6A) and although some genera were detected only by a single platform (26 and 28 genera by Illumina and Nanopore, respectively), these groups had very low abundance levels (<1.4%). In total, 16 genus-level biomarkers were identified for each of the four months by both platforms (Figure 6B), with only four detected exclusively by one platform (one by Illumina and three by Nanopore).

4 | Discussion

Understanding the role of sea lice in the transmission of salmon pathogens is critical for effective disease management in Atlantic salmon aquaculture (Haydon et al. 2002; Wilson et al. 2020). In this study, we characterised the gut bacterial microbiome of *Lepeophtheirus salmonis* collected from a commercial marine salmon farm in Ireland. The gut bacterial microbiome was dominated by members of the phyla Pseudomonadota and Bacteroidota, with clear temporal shifts observed over the four-month sampling period, primarily reflecting increased diversity. Across all samples, 24 pathogen-associated genera were identified, including 15 putative pathogens. Despite differences in sequencing technology, analyses based on both Illumina MiSeq and Oxford Nanopore platforms yielded consistent and comparable results, supporting the robustness of our findings.

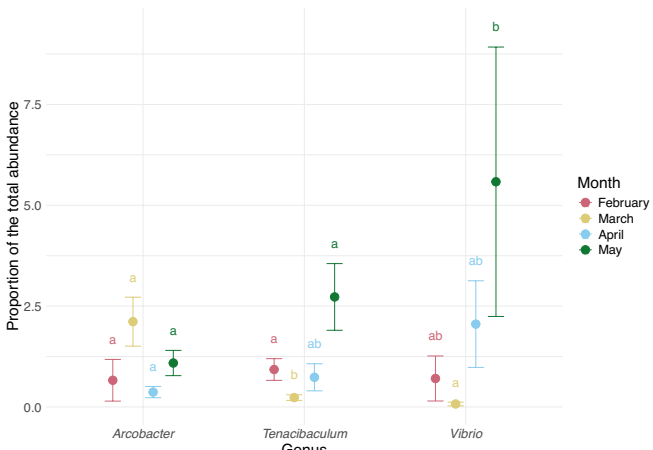


FIGURE 4 | Temporal changes in sea lice gut microbiome pathogen-associated genera, *Arcobacter*, *Tenacibaculum*, and *Vibrio*, over a four-month period. The mean ± SE of abundance for each genus is shown for each month. Statistical differences between months are represented by lower case letters.

4.1 | Composition of Sea Lice Gut Bacterial Microbiome

The gut bacterial microbiome of *L. salmonis* collected from Irish salmon farms was comparable to previous descriptions of sea lice

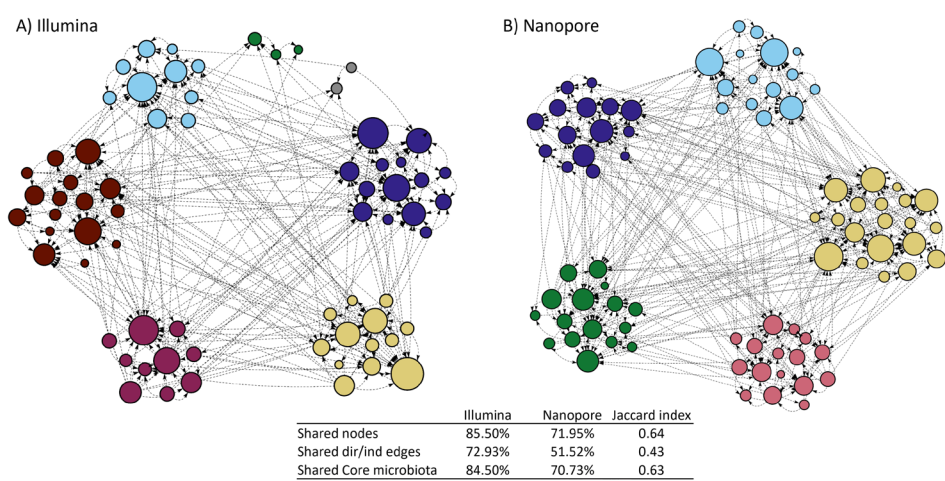


FIGURE 5 | Bayesian networks representing sea lice gut microbiome models inferred from read counts obtained using (A) Nanopore and (B) Illumina sequencing platforms. Circles represent bacterial taxa, and their size reflects differences in the relative abundance normalised within each model. The organisation in different clusters (7 for Illumina and 5 for Nanopore data; indicated with different colours) was obtained through the application of the Leiden community detection algorithm in order to visualise the densely connected nodes.

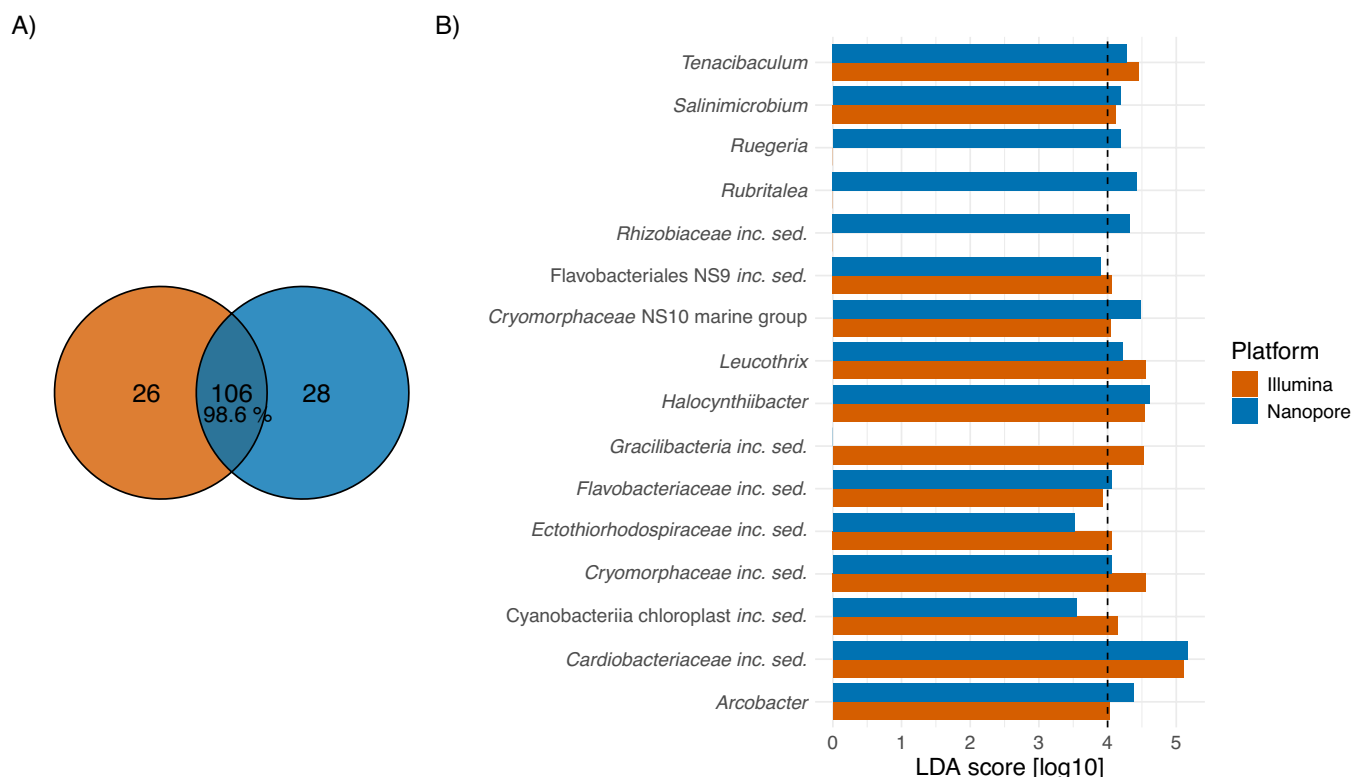


FIGURE 6 | (A) Venn diagram showing the differences in detected genera within the sea lice gut microbiome for Illumina and Nanopore sequencing platform with an overall abundance (%) of shared genera. (B) LDA scores of genus-level biomarkers identified for each sequencing platform; the dashed line denotes the threshold (4) for the LDA score.

microbiota (Dalvin et al. 2020; Gonçalves et al. 2020). The community was dominated by Pseudomonadota (62%–91%) and, to a lesser extent, Bacteroidota (9%–17%). These proportions closely mirror those reported by Gonçalves et al. (2020), who observed >60% Pseudomonadota and ~10% Bacteroidota in *C. rogercresseyi*. At the class level, Gammaproteobacteria, Flavobacteria, and Alphaproteobacteria were consistently among the most abundant groups, again in agreement with previous findings (Gonçalves et al. 2020). In contrast, Dalvin et al. (2020) identified Flavobacteriales (30%–71%, Bacteroidota) and Burkholderiales (6%–37%, Pseudomonadota) as dominant taxa in *L. salmonis*, highlighting potential differences between host populations and environmental conditions. Dalvin et al. (2020) also reported low relative abundance (1%–16%) of Spirochaetales and Mycoplasmales, along with sporadic detection of Pseudomonadales, Pseudoalteromonadales, Alteromonadales, Aeromonadales, Vibrionales, and Rickettsiales. Of these groups, only Pseudomonadales and Rickettsiales were detected in the present study.

The three studies published to date on the gut bacterial microbiome of sea lice employed different methodological approaches. Dalvin et al. (2020) used whole-metagenome shotgun sequencing, whereas Gonçalves et al. (2020) and Morales-Rivera et al. (2022) focused on Nanopore sequencing of the full-length 16S rRNA gene, which is comparable to the Nanopore workflow applied in the present study. However, differences also exist in the DNA extraction kits and bioinformatics pipelines used across these studies (Dalvin et al. 2020; Gonçalves et al. 2020; Morales-Rivera et al. 2022).

The composition of the sea lice gut bacterial microbiome is likely shaped by multiple factors, including the microbial communities present in salmon mucus and skin, which constitute the primary food source of the lice, as well as those in the surrounding water environment (Llewellyn et al. 2017; Gonçalves et al. 2020). However, the salmon mucosal microbiome itself is dynamic and strongly influenced by diet, salinity, and parasite load (Landeira-Dabarca et al. 2013; Lokesh and Kiron 2016; Llewellyn et al. 2017). For *C. rogercresseyi*, Gonçalves et al. (2020) identified a stable core gut bacterial microbiome that, while comprised of only 19% distinct taxa, accounted for 70% total abundance across regions and farming practices. Similarly, in the present study, most genera detected in *L. salmonis* formed a stable core (80% and 84% distinct taxa for Illumina and Nanopore, respectively), accounting for 99% overall abundance. Nevertheless, as samples were restricted to a single farm over a four-month period, it remains unclear whether the observed core gut bacterial microbiome is broadly representative of *L. salmonis* populations or specific to the aquaculture site during the four-month period studied. Moreover, a direct comparison to the salmon mucosal microbiome was not possible, as host-associated samples were not included in this study.

Temporal dynamics of sea lice gut bacterial microbiomes remain poorly understood (Moisander et al. 2015). Previous work suggests that microbial associations in copepods can be transient or persistent, depending on host life stage, moulting events, physiological state, and environmental conditions (Moisander et al. 2015; Datta et al. 2018; Bass et al. 2021). For example, microbial colonisation of the exoskeleton may reset

after moulting, with subsequent shifts reflecting recolonisation dynamics (Moisander et al. 2015). Environmental drivers, including rising temperatures, may further influence gut bacterial microbiome composition by accelerating bacterial growth rates (Heidelberg et al. 2002; Moisander et al. 2015). In this study, short-term temporal variation in the *L. salmonis* gut bacterial microbiome was mainly reflected in diversity patterns. Microbial diversity increased over the sampling period, peaking in April and reaching its lowest point in March. This pattern appeared to be driven largely by a decline in the relative abundance of Cardiobacteriales, allowing other taxa to expand and thereby enhancing overall community evenness. Such shifts could be driven by multiple, interacting factors, including management interventions affecting the salmon skin microbiome and seasonal increases in seawater temperature (Klimesova et al. 2025). Further studies incorporating longer time series and environmental metadata are needed to clarify the relative roles of environmental factors and management practices in shaping temporal changes in sea lice gut bacterial microbiome.

4.2 | Identification of Potential Fish Pathogens

For fish ectoparasites to function as effective pathogen vectors, they must be capable of moving between hosts, breaching the epidermis, and transmitting infectious agents (Cusack and Cone 1986). The species *L. salmonis* meets these criteria through its mobility, invasive feeding behaviour, and demonstrated ability to harbour fish pathogens (Novak et al. 2016). In addition, it can suppress host inflammatory responses and modulate immune function (Fast et al. 2007; Lewis et al. 2014), potentially enhancing pathogen transmission (Barker et al. 2019). However, species of *Caligus*, which infect a wider range of hosts, may represent an even greater risk as pathogen vectors than *L. salmonis* (Hemmingsen et al. 2020).

Although three studies have characterised the sea lice gut bacterial microbiome (Dalvin et al. 2020; Gonçalves et al. 2020; Morales-Rivera et al. 2022), the role of sea lice in pathogen transmission remains only partially understood (Nylund et al. 1991; Jakob et al. 2011; Novak et al. 2016). The sea louse *C. rogercresseyi* was confirmed to act as a mechanical vector for *Piscirickettsia salmonis* (Labra et al. 2020) and infectious salmon anaemia virus (ISAV) (Oelckers et al. 2014). Other pathogens that were detected in the *C. rogercresseyi* gut bacterial microbiome and for which sea lice have the potential to act as vectors are: *Tenacibaculum* (*T. maritimum*, *T. ovolyticum*, *T. discolor*, *T. dicentrarchi*, *T. gallaicum*, *T. saleae*), *Vibrio* (*V. splendidus*), *Pseudoalteromonas* and *Moritella* (*M. viscosa*) (Gonçalves et al. 2020; Morales-Rivera et al. 2022).

In comparison, *L. salmonis* has been confirmed as a mechanical vector of infectious haematopoietic necrosis virus (IHNV) (Jakob et al. 2011) and a vector of *Aeromonas salmonicida* under laboratory conditions, although it remains unclear whether transmission occurs mechanically or biologically (Novak et al. 2016). Pathogens previously detected in *L. salmonis* include *A. salmonicida* (Nese and Enger 1993; Novak et al. 2016), *T. maritimum*, *P. fluorescens*, *V. alginolyticus*, *V. vulnificus* (Barker et al. 2009), *Desmozoon lepeophtheirii* (Freeman

and Sommerville 2009) or *Paranucleospora theridion* (Nylund et al. 2010; Økland 2012), salmonid alphavirus (Pettersson et al. 2009), IHNV (Jakob et al. 2011), infectious pancreatic necrosis virus (Johnson et al. 2004) and *Neoparamoeba perurans* (Nowak et al. 2010). Detection of *N. perurans* in sea lice, however, has been suggested to be incidental and not necessarily indicative of a transmission role (Nowak et al. 2010).

In the present study, 24 pathogen-associated genera and a total of 15 putative pathogenic species were identified in the gut bacterial microbiome of sea lice collected from Irish salmon farms. Of these, nine species (*A. johnsonii*, *A. sobria*, *A. logei*, *A. wodanis*, *M. marina*, *P. fluorescens*, *T. dicentrarchi*, *T. maritimum*, and *V. anguillarum*) have previously been reported from salmonid hosts (Austin and Austin 2016; Avendaño-Herrera et al. 2016; Lim and Hong 2020). In addition, *V. splendidus* and *V. tapetis* have been described as pathogens causing mortality in corksing wrasse (*Symphodus melops*) (Jensen et al. 2003; Treasurer 2012). Among the detected pathogens, *T. maritimum*, *T. dicentrarchi*, and *V. anguillarum* are of particular concern due to their substantial economic impact on aquaculture as the causative agents of tenacibaculosis and vibriosis in Atlantic salmon, respectively (Avendaño-Herrera et al. 2006; Frans et al. 2011; ASTRAL 2023). The identified pathogens in the sea lice gut bacterial microbiome are consistent with previous reports (Barker et al. 2009; Morales-Rivera et al. 2022) and further expand the diversity of known pathogens associated with the sea lice gut bacterial microbiome. Identification of these pathogens is an important first step in the determination of the role that sea lice play in transmission.

Although most pathogen-associated genera were present at low abundance levels (<0.6%), *Arcobacter*, *Tenacibaculum*, and *Vibrio* were consistently detected across all samples at relatively higher abundances (1%–2%). Temporal variation was evident in *Tenacibaculum* and *Vibrio*, with the lowest abundances recorded in March and the highest in May. The increase in the abundance of pathogen-associated genera mirrored the rise in overall microbial diversity observed in the sea lice gut bacterial microbiome. However, due to the lack of detailed data on disease prevalence at the salmon farm, it was not possible to determine whether the increased abundance of these genera in sea lice was associated with higher disease incidence in the farmed salmon population.

Although a wide variety of putative salmon pathogens were detected in the sea lice gut bacterial microbiome, their relationship to disease in Atlantic salmon remains unclear. As the analysis detected only pathogen DNA, it cannot be determined whether the bacteria were viable (Keer and Birch 2003). Furthermore, the detected DNA may have originated from the fish surface or the surrounding environment rather than from active infection (Llewellyn et al. 2017). Even if pathogens were present in infected fish, their abundance may have been too low to cause observable disease or trigger an outbreak (Boerlage et al. 2017). These findings highlight the complexity of vector–host–pathogen interactions and underscore the need for further research to clarify the role of sea lice in the transmission and life cycle of salmon pathogens. Future research should prioritize examining sea lice during disease outbreaks, isolating and culturing bacteria from sea lice, and conducting experimental infection studies.

4.3 | Comparison of Two Sequencing Platforms

Comparisons of Illumina and Nanopore sequencing platforms consistently show that the choice of platform, along with methodological and analytical factors, can lead to some differences in reported bacterial profiles, diversity estimates, and community composition (Yeo et al. 2024; Hoffbeck et al. 2025). While the two methods are broadly concordant at higher taxonomic levels, agreement weakens at lower levels (Yeo et al. 2024), where Nanopore's long-read capability provides advantages for pathogen detection and the identification of unique taxa, though primer mismatches may limit detection of certain genera (Nygaard et al. 2020; Esberg et al. 2024; Yeo et al. 2024). These findings highlight the need for caution when interpreting absolute measures of diversity across platforms and suggest that relative comparisons within studies may be more reliable (Hoffbeck et al. 2025).

Rarefaction curves for both platforms reached an asymptote, indicating sufficient sequencing depth. Although Nanopore generated more reads on average (199,578 vs. 81,125), comparison at an equivalent depth (~80,000 reads) showed that it recovered a similar or higher number of distinct genera than Illumina, which was also reflected in the higher average genera richness (Observed richness and ACE) of the Nanopore dataset. However, richness metrics displayed divergent temporal patterns between platforms, with month-to-month trends not aligning, whereas diversity indices exhibited consistent temporal variation in both datasets. Such discrepancies in richness have been previously documented, as richness estimators are particularly sensitive to sequencing errors, read-length differences and pipeline-specific denoising thresholds (Stevens et al. 2023; Szoboszlay et al. 2023). In contrast, diversity indices tend to be more robust and often show consistent ecological patterns across methods (Heikema et al. 2020; Stoeck et al. 2024). Diversity estimates can also be influenced by the approach employed (Hoffbeck et al. 2025), and in this dataset, Illumina samples showed higher average diversity (Shannon and Simpson index) than Nanopore. However, despite differences in absolute values, both platforms revealed comparable seasonal fluctuations, with a minimum in March and a peak in April.

Beyond diversity and richness, both platforms produced highly comparable gut bacterial microbiome compositions. The same dominant taxonomic groups were detected, although relative abundances differed slightly. As expected, discrepancies became more apparent at lower taxonomic levels, consistent with observations by Yeo et al. (2024), who reported stronger correlations at the phylum level (median 0.76) than at the genus level (median 0.68). In total, 26 and 28 genera were uniquely detected by Illumina and Nanopore sequencing, respectively, but together these accounted for only 1.4% of total community abundance, indicating that platform-specific differences were largely restricted to low-abundance taxa. Importantly, both platforms captured similar seasonal dynamics in community composition, with greater similarity between February and March and between April and May.

The long-read capability of Nanopore sequencing enabled species-level taxonomic resolution, allowing the identification

of 15 specific pathogens within the pathogen-associated genera. This enhanced resolution, consistent with previous reports (Nygaard et al. 2020; Yeo et al. 2024), highlights the advantage of Nanopore sequencing for pathogen detection. Although concerns have been raised regarding the accuracy of Nanopore at the species level, improvements in Nanopore chemistry, combined with optimised sequencing and bioinformatic pipelines, have been shown to produce robust and reliable results (Benítez-Páez et al. 2016; Bertolo et al. 2024). Along with its relatively low cost, these characteristics underscore its potential as an efficient tool for rapid pathogen screening (Esberg et al. 2024; Yeo et al. 2024).

The Bayesian network analysis provided further evidence of consistency between the two sequencing platforms. Structural differences between the Nanopore- and Illumina-based networks, particularly in clustering patterns, likely reflect differences in sequencing depth and read distribution. Nevertheless, the strong overlap in core bacterial nodes and the comparable agreement across taxa associations indicate that the two platforms reconstructed analogous biological relationships, supporting the robustness of the inferred microbial network. In parallel, LEfSe analysis revealed largely comparable results between the two platforms, with 16 shared genus-level biomarkers and one and three detected exclusively by Illumina and Nanopore, respectively. Non-platform-specific markers exhibited highly similar LDA effect sizes, indicating that enrichment signals were comparable when a marker was detected by both approaches.

In summary, the comparison of Illumina and Nanopore demonstrates that, despite technical and methodological differences, both platforms provided broadly consistent insights into the sea lice gut bacterial microbiome. Nanopore offered advantages in taxonomic resolution and pathogen detection. Importantly, both approaches revealed comparable community composition, temporal trends, and network structures, underscoring the robustness of the observed biological patterns and supporting the reliability of conclusions drawn across sequencing methodologies.

5 | Conclusions

This study provides the first detailed characterisation of the sea lice gut bacterial microbiome on an Irish salmon farm, revealing a diverse bacterial community that includes several known fish pathogens. Despite methodological differences, Illumina and Nanopore sequencing platforms produced broadly consistent biological interpretations, with Nanopore offering the additional advantages of species-level resolution and rapid pathogen detection. These findings represent an important first step toward understanding the role of sea lice in pathogen dynamics within salmon aquaculture systems. Nevertheless, further research is needed to determine pathogen survival, transmission to hosts, and links to disease in salmon populations. Collectively, the results pave the way for developing non-invasive pathogen surveillance strategies in Atlantic salmon aquaculture through analysis of the sea lice gut bacterial microbiome.

Author Contributions

B.K.: conceptualisation, methodology, software, validation, formal analysis, data curation, writing – original draft, writing – review and editing, visualisation, funding acquisition. N.M.R.: resources, conceptualisation, writing – review and editing; R.D.-B.: methodology, software, validation, formal analysis, investigation, data curation, writing – review and editing. F.M.: software, validation, formal analysis, writing – review and editing. F.N.-C.: validation, writing – review and editing. J.P.-S.: methodology, software, validation, resources, writing – review and editing, supervision. K.O.: conceptualisation, writing – review and editing. O.L.: writing – review and editing. H.R.: writing – review and editing. A.T.: conceptualisation, methodology, validation, resources, writing – original draft, writing – review and editing, supervision, project administration, funding acquisition. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used in this study are available at <https://doi.org/10.6084/m9.figshare.30836501>. Raw sequence data are available on Sequence Read Archives under the Bioproject accession number: PRJNA1372954 (<https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA1372954>). The R code used in this study is available on GitHub at https://github.com/Klimesobe/Sealice_Microbiome.git.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** jfd70182-sup-0001-Supinfo1.docx.